

Recall — data handling, and the project topic

Tally, frequency, the three averages and the range — and the question the project will answer.

LESSON 157

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 6 COLLECTING DATA

LESSON CLOCK

8'

12'

12'

8'

Collecting and recording	8 min
The three averages and the range	12 min
The mean from a frequency table	12 min
The project topic	8 min

LEARNING OBJECTIVES

- Record data in a tally and turn it into a frequency table.
- Find the mean, the median, the mode and the range of a small set.
- Find the mean from a frequency table without listing every value.
- Frame a question that a survey can actually answer.

TEXTBOOK

National	pp. 449–452
Cambridge	Stage 7, pp. 58–62
Workbook	Exercise 6.1

01

Explanation

Collecting and recording

Data is either a category — a favourite sport, a colour — or a number. Numbers that can only be counted, such as marks out of ten, are **discrete**; those that are measured, such as heights, are **continuous**.

MARK	TALLY (A CROSSED GROUP IS FIVE)	FREQUENCY
4		1
5		2
6		3
7	/	5
8		4
9		3
10		2

THE FREQUENCIES MUST ADD TO THE NUMBER OF PEOPLE ASKED

Here $1 + 2 + 3 + 5 + 4 + 3 + 2 = 20$. That single check catches most recording errors before any calculation begins.

The three averages and the range

MEASURE	WHAT IT IS	FOR 2, 5, 5, 6, 12
mean	the total shared out equally	$30 \div 5 = 6$
median	the middle value, in order	5
mode	the value that occurs most	5
range	largest minus smallest	$12 - 2 = 10$

THE RANGE IS NOT AN AVERAGE

It says how spread out the data is, not where its centre lies. Two classes can share a mean of 6 and have ranges of 2 and 10 — very different classes.

For an even number of values the median is halfway between the two middle ones.
Order the data first, every time; an unordered list gives the wrong median more often than not.

The mean from a frequency table

Twenty marks need not be written out. Multiply each mark by its frequency, add, and divide by the total frequency.

MARK X	FREQUENCY F	$X \cdot F$
4	1	4
5	2	10
6	3	18
7	5	35
8	4	32
9	3	27
10	2	20
total	20	146

$$\text{mean} = \frac{146}{20} = 7.3$$

MODE, MEDIAN AND RANGE COME OFF THE SAME TABLE

The mode is the mark with the largest frequency — 7. The median is between the 10th and 11th values, both of which are 7. The range is $10 - 4 = 6$.

The project topic

THE PROJECT, ANNOUNCED TODAY AND PRESENTED IN LESSON 203

Which sport do the pupils of our school prefer? Ask at least 60 pupils, record the answers in a tally, and present the result as a pie chart with the angle of every sector shown.

STAGE	LESSONS	WHAT IS PRODUCED
plan the survey	158–159	the question and the sample
collect and tally	at home	a frequency table
draw the chart	160–163	a pie chart with angles
interpret it	164–166	three written conclusions
present	203–204	a poster and four minutes of talk

A GOOD QUESTION HAS A SMALL, FIXED LIST OF ANSWERS

“Which sport do you prefer?” with six options can be tallied. “What do you think about sport?” cannot. Deciding the options before asking anyone is half the work of a survey.

02

Worked examples

EXAMPLE 1 Find the mean, median, mode and range of 2, 5, 5, 6, 12.

1

Total 30, so the mean is $30 \div 5 = 6$.

2

In order, the middle value is 5.
Already in order ✓

3

The mode is 5; the range is $12 - 2 = 10$.
Three different “middles”.

ANSWER

Mean 6, median 5, mode 5, range 10

EXAMPLE 2 Find the mean of the twenty marks in the frequency table.

- ① Multiply each mark by its frequency: 4, 10, 18, 35, 32, 27, 20.

- ② Total 146 over 20 pupils.

- ③ $146 \div 20 = 7.3$.
Between 4 and 10 ✓

ANSWER

7.3

EXAMPLE 3 From the same table, find the median and the mode.

- ① With 20 values the median is between the 10th and 11th.

- ② Cumulative frequencies: 1, 3, 6, 11 — both fall on 7.

- ③ The mode is 7, with frequency 5.
Median and mode agree here.

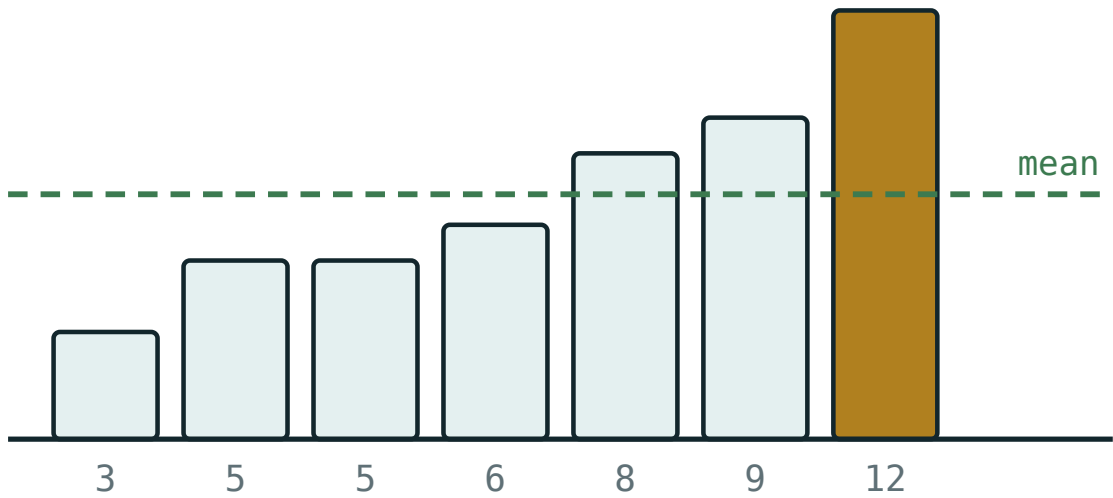
ANSWER

Median 7, mode 7

03 Interactive model

Drag the largest value far to the right in the model: the mean chases it, the median does not. That is the whole argument for the median in one movement.

Mean, median, mode and range



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Data	Ma'lumot	Данные
Tally	Sanoq belgisi	Подсчёт чёрточками
Frequency	Chastota	Частота
Frequency table	Chastotalar jadvali	Таблица частот
Mean	O'rtacha arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Mode	Moda	Мода
Range	Farq (kenglik)	Размах

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The mode is:

the middle value

the most common value

the total $\div$ the count

the spread

The median of 2, 5, 5, 6, 12 is:

5

6

5.5

10

The range of that set is:

5

6

10

12

The mean of the twenty marks is:

7

7.3

146

20

Before finding a median you must:

add the values

order the values

count the modes

draw a chart

Heights of pupils are:

categorical

discrete

continuous

a tally

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The mode of 2, 5, 5, 6, 12

ANSWER 5

2. The median of 2, 5, 5, 6, 12

ANSWER 5

3. The mean of 2, 5, 5, 6, 12

ANSWER 6

4. The range of 2, 5, 5, 6, 12

ANSWER 10

5. The mean of 3, 4, 4, 5, 9

ANSWER 5

6. The mode of the twenty marks

ANSWER 7

7. The number of pupils in the table

ANSWER 20

Medium · the standard question

7 PROBLEMS

1. The median of the twenty marks

ANSWER 7

2. The mean of the twenty marks

ANSWER 7.3

3. The range of the twenty marks

ANSWER 6

4. The total of the twenty marks

ANSWER 146

5. The frequency of the mark 8

ANSWER 4

6. How many scored 8 or more

ANSWER 9

7. How many scored below 7

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. A tally of four groups of five and two more

ANSWER 22

2. Which average is least affected by one very large value?

ANSWER The median

3. A 21st pupil scores 5: the new mean, to 1 d.p.

ANSWER 7.2

4. If every mark rose by 1, the new mean

ANSWER 8.3

5. ...and the new range

ANSWER 6, unchanged

6. Which average must always be one of the data values?

ANSWER The mode

7. Data that is measured rather than counted is called

ANSWER Continuous

07 Homework

Homework — 5 tasks

Bring the frequency table of your own survey to the next lesson; the project depends on it.

1. Find the mean, median, mode and range of 3, 7, 7, 8, 10, 13.
2. A frequency table has marks 5, 6, 7, 8 with frequencies 2, 5, 8, 5. Find the mean.
3. Find the median and the mode of that table.
4. Write your survey question with a list of at most six possible answers.
5. Explain in one sentence why the range is not an average.